

Research Article

Cooperative Assembly Drives Buried Interface Engineering for Efficient Inverted Perovskite Solar Cells

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Materials and Performance Parameters of the Champion Cell	
Photoactive material(s) used	
Please provide the values for power conversion efficiency (PCE [%]), open circuit voltage (V_{oc} [mV]), short circuit current density (J_{sc} [mA/cm ²]), and Fill Factor (FF[%]).	
Is stabilized performance reported (please see ref.1 for details)?	
Are current density-voltage (J-V) plots provided in both forward and backward directions (only applicable for perovskite solar cells)?	Yes No
J-V Measurement of the Champion Cell	
Please state whether your devices were tested under standard test conditions (T_{cell} : 25°C; J-V incident light intensity: 100 mW/cm ²).	Yes No
Please state the reference spectrum used.	
Was a spectrally adjustable solar simulator used (only applicable for multijunction cells)?	Yes No
Mask aperture area [cm ²]	
Total device active area [cm ²]	
J-V measurement data file* Inclusion of the J-V plot in the manuscript is mandatory; please indicate whether you have also uploaded* the measurement data file (encouraged)	Yes No
Bandgap Energy Measurement of the Champion Cell	
Bandgap Energy E_g [eV] (taken from the EQE spectrum, at the inflection point of the absorption threshold) (please see ref.2 for details).	

Integrated J _{sc} [mA/cm ²] (taken from the EQE spectrum under 1 sun AM1.5G illumination)	
EQE data file* Inclusion of the EQE spectrum in the manuscript is mandatory; please indicate whether you have also uploaded* the data file (encouraged)	Yes No
Semitransparent PV (only applicable for transparent/semitransparent devices)	
Average visible transmittance, AVT [%] (please see 'Data S1' in ref.3 for the calculation)	
Transmittance measurement data file* Inclusion of the transmittance spectrum in the manuscript is mandatory for transparent/semitransparent devices; please indicate whether you have also uploaded* the data file (encouraged)	Yes No
Have you confirmed EQE[%]+Transmittance [%]<1?	Yes No
Long-Term Stability Test	
Please provide the initial values for PCE [%], V _{oc} [mV], J _{sc} [mA/cm ²], and FF[%]	
Please state incident light spectrum [e.g., AM1.5G, white LED, etc.], incident illumination intensity (e.g., 1 sun), electrical bias condition (open circuit/MPP/short circuit/constant load), UV filter, test temperature, cell encapsulation, atmosphere composition (air/N ₂ /Ar), relative humidity [%], etc. (please see 'Table 3' in ref.4 for more details).	

* to be submitted as a separate file when uploading your production data:

J-V measurement data file: two columns csv format: Voltage [V], Current density [mA/cm²]

EQE data file: two columns csv format: wavelength [nm], EQE [%]

Transmittance measurement data file: two columns csv format: wavelength [nm], T [%]

1. Dunbar, R. B. *et al.* [How reliable are efficiency measurements of perovskite solar cells? The first inter-comparison, between two accredited and eight non-accredited laboratories.](#) *J. Mater. Chem. A*, **2017**, 5, 22542.
2. Almora, O. *et al.* [Quantifying the Absorption Onset in the Quantum Efficiency of Emerging Photovoltaic Devices.](#) *Adv. Energy Mater.* **2021**, 11, 2100022.
3. Yang, C. *et al.* [How to accurately report transparent solar cells.](#) *Joule*, **2019**, 3, 8, 1803-1809.
4. Khenkin, M.V., Katz, E. A., Abate, A. *et al.* [Consensus statement for stability assessment and reporting for perovskite photovoltaics based on ISOS procedures.](#) *Nat. Energy*, **2020**, 5, 35– 49.

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Cooperative Assembly Drives Buried Interface Engineering for Efficient Inverted Perovskite Solar Cells

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Keywords: perovskite solar cells, self-assembled monolayers, synergistic assembly, perovskite module

29

Abstract: Self-assembled monolayers (SAMs) containing carbazole and phosphonic acid groups have emerged as effective hole-transport layers (HTLs) for high-performance inverted perovskite solar cells (PSCs). However, the intrinsic aggregation tendency of SAM molecules often compromises interfacial uniformity, inducing non-radiative recombination and limiting device efficiency and stability. Here, we report a synergistically assembled chemically heterogeneous (SACH) interfacial layer constructed by integrating [4-(3,6-dimethyl-9H-

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1 carbazol-9-yl)butyl]phosphonic acid (Me-4PACz) with carbazole-9-ethanol (CE). In this
 2 strategy, CE induces cooperative π - π interactions and van der Waals coupling with Me-4PACz,
 3 transforming conventional competitive co-adsorption into an ordered molecular co-assembly
 4 process. This suppresses the self-aggregation of Me-4PACz, enabling a more uniform HTL,
 5 and enhancing charge extraction. Meanwhile, the hydroxyl group of CE forms O-H $\cdots$ I-
 6 interactions with perovskite, passivating iodine vacancies and uncoordinated Pb/I defects,
 7 thereby improving crystallinity and reducing buried-interface recombination. As a result,
 8 devices employing the SACH layer achieve a PCE of 27.06% (certified 26.86%) alongside
 9 excellent stability. Under maximum power point tracking in ambient air at 65 °C, encapsulated
 10 SACH-based devices retained 88.6% of their initial efficiency after 500 h. Large-area
 11 perovskite modules (30 $\times$ 30 cm²) further demonstrate promising performance with an
 12 efficiency of 21.90%, highlighting the broad applicability of this cooperative interfacial design
 13 for industrial-scale perovskite photovoltaics.

14

15 1. Introduction

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17 Metal halide perovskite solar cells (PSCs) have rapidly advanced over the past decade,
 18 propelled by their high defect tolerance, strong absorption coefficient, tunable bandgap, and
 19 exceptional power conversion efficiencies (PCEs), recently surpassing 27%^[1-5]. Inverted (p-i-
 20 n) device architectures, in particular, have gained increasing prominence due to their suppressed
 21 hysteresis, enhanced device robustness, and compatibility with scalable fabrication pathways^[6].
 22 Within such architectures, further efficiency gains are increasingly constrained by losses
 23 originating from the buried perovskite/HTL interface, underscoring the need for precise
 24 molecular-level control over interfacial chemistry and structure^[7].

25 Self-assembled monolayers (SAMs), specifically those derived from carbazole
 26 phosphonic acids such as [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Me-
 27 4PACz), have emerged as outstanding HTLs owing to their strong anchoring ability, tunable
 28 energy levels, and ability to form ultrathin, low-loss charge-extraction layers^[8-11]. However, the
 29 amphiphilic nature of Me-4PACz leads to molecular self-aggregation, rendering the resulting
 30 monolayers structurally heterogeneous^[12-13]. These morphological irregularities hinder efficient
 31 charge extraction and disrupt uniform perovskite nucleation, giving rise to interfacial
 32 recombination losses^[14-16]. To mitigate these limitations, both physical and chemical interfacial
 33 engineering strategies have been pursued. Metal oxide overlayers (e.g., Al₂O₃^[17]) can improve
 34 surface wettability yet fail to fully address molecular-scale disorder. Multicomponent co-
 35 adsorbed SAM systems offer a chemical route to modulate interfacial energetics and defect
 36 passivation^[18-20], but traditional co-adsorption inherently suffers from competitive binding

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among molecules with distinct anchoring groups, steric features, and diffusion dynamics^[21]. This competitive adsorption complicates the formation of ordered monolayers and hinders precise control over interfacial chemical composition. In all, despite their widespread use, the intrinsic aggregation behavior of carbazole-based SAMs remains insufficiently addressed, representing a fundamental bottleneck for achieving structurally ordered and electronically optimal interfaces.

Here, we introduce a synergistically assembled chemically heterogeneous (SACH) interface that transforms competitive co-adsorption into cooperative co-assembly through the incorporation of carbazole-9-ethanol (CE). CE lacks a phosphonic acid anchoring group and therefore avoids binding-site competition with Me-4PACz; instead, CE engages in favorable π - π stacking and van der Waals interactions with Me-4PACz, guiding the latter into a more ordered monolayer arrangement. This process suppresses self-aggregation while simultaneously imparting additional defect passivation functionality through CE's hydroxyl group. The resulting buried interface exhibits enhanced wettability, stronger interfacial coupling, improved perovskite crystallinity, and reduced non-radiative recombination, enabling PSCs with a champion PCE of 27.06% (certified 26.86%) and excellent operational stability. Moreover, $30 \times 30 \text{ cm}^2$ perovskite modules fabricated via the SACH strategy reach an efficiency of 21.90%, highlighting its promise for scalable and industrially viable photovoltaic applications.

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21 **2. Results and Discussion**

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Given the critical role of interfacial wettability in perovskite nucleation and film continuity, we first examined the impact of CE incorporation on the surface properties of the HTL (Figure S1). The perovskite film deposited on pure Me-4PACz exhibited poor surface coverage, attributed to the intrinsically hydrophobic surface chemistry and aggregation tendency of Me-4PACz. By contrast, CE incorporation led to nearly void-free perovskite films with uniform surface coverage, indicative of substantially enhanced substrate hydrophilicity, as further corroborated by the reduced contact angle (Figure S2).

The origin of this improvement could be attributed to the cooperative intermolecular interactions between CE and Me-4PACz. The molecular design of CE was guided by eliminating anchoring-site competition while preserving strong intermolecular affinity with Me-4PACz. As illustrated in Figure 1a, CE retains the carbazole core of Me-4PACz but replaces the phosphonic acid group with a hydroxyl moiety. Crucially, this hydroxyl group not only avoids direct surface binding competition but also contributes to the improved hydrophilicity

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observed above. Electrostatic potential (ESP) mapping reveals a pronounced intramolecular dipole within CE, providing complementary hydrogen-bonding and van der Waals interaction sites (Figure 1b), while the electron-rich carbazole π -core enables favorable π - π stacking with Me-4PACz. Density functional theory (DFT) calculations substantiate these design principles, showing that CE exhibits a stronger adsorption affinity toward Me-4PACz (-0.27 eV) than Me-4PACz self-association (-0.20 eV). This energy preference suggests a thermodynamic driving force for CE to intercalate between adjacent Me-4PACz molecules, thereby suppressing self-aggregation and promoting ordered face-to-face π - π stacking configurations (Figure 1c). FTIR spectra (Figure 1d) further verify these interactions: compared with the pure CE and Me-4PACz solutions, the mixed Me-4PACz+CE system shows blue-shifted C=C/C=N stretching bands (1590.0 cm^{-1} and 1320.5 cm^{-1}), consistent with strengthened intermolecular coupling.

Specifically, rather than forming randomly distributed molecular clusters, the Me-4PACz molecules undergo a more controlled assembly process. The cooperative interactions effectively weaken the homomolecular affinity, allowing the molecules to diffuse and distribute evenly across the substrate. Consequently, the phosphonic acid anchoring groups can bind uniformly to the surface, forcing the functional end-groups into a highly ordered, vertically aligned orientation. This dynamic transition suppresses localized agglomeration, yielding a smooth, densely packed monolayer with distinct chemical heterogeneity at the nanoscale, as schematically illustrated in Figure 1e. Atomic force microscopy (AFM) analysis confirms that NiO_x/SACH layer displays a more uniform surface morphology with reduced root-mean-square (RMS) roughness (23.5 nm vs. 24.8 nm for Me-4PACz alone, Figure S3). These results demonstrate that CE promotes a more ordered packing configuration, in sharp contrast to the aggregation-dominated, cluster-like distributions typically observed in single-component Me-4PACz SAMs.

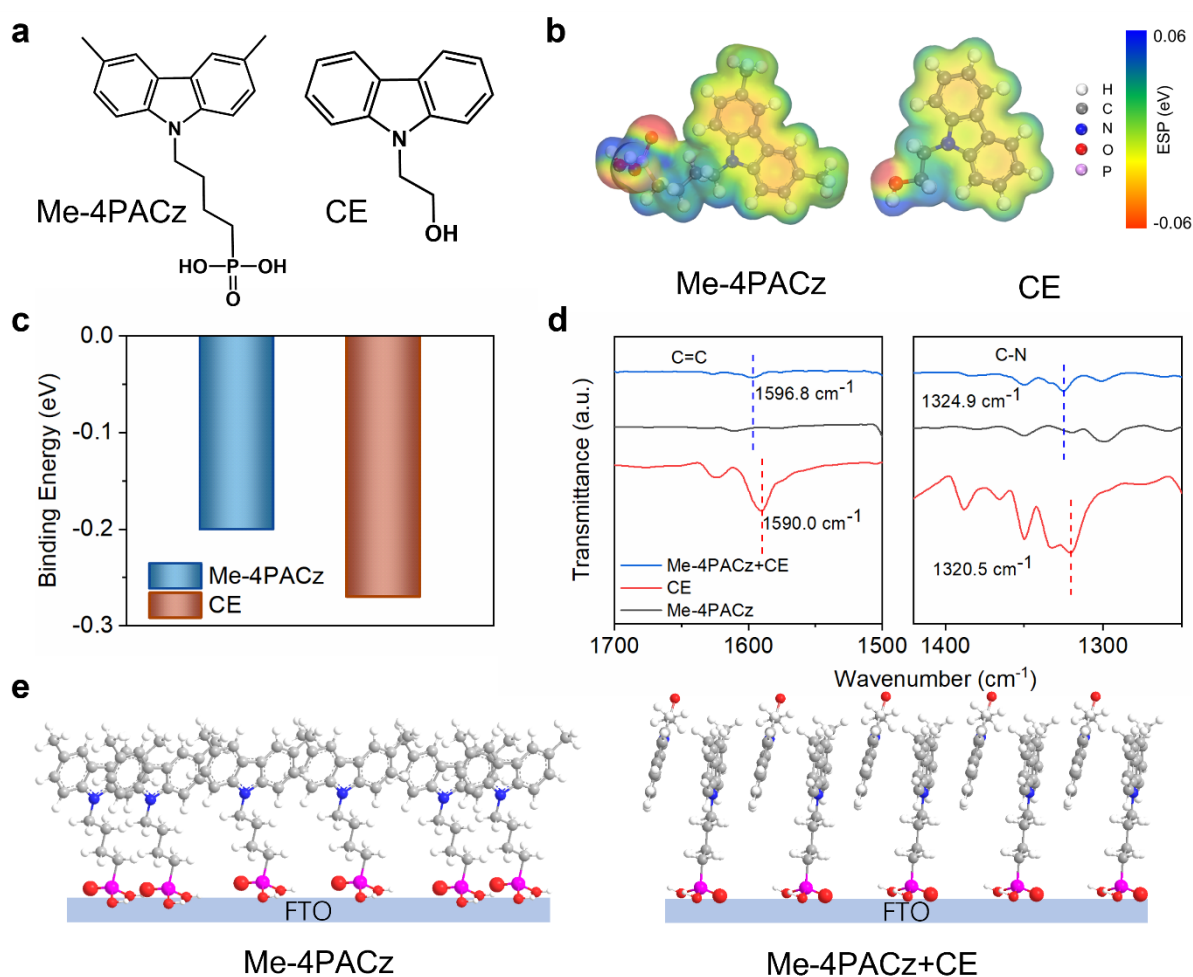


Figure 1. The interaction analysis of Me-4PACz and CE. (a) Molecular structures of Me-4PACz and CE. (b) ESP of Me-4PACz and CE. (c) DFT calculation of the binding energy of Me-4PACz and CE with Me-4PACz. (d) FTIR spectra of Me-4PACz, CE, Me-4PACz+CE. (e) Schematic illustration of the molecular assembly of Me-4PACz and CE on FTO substrate.

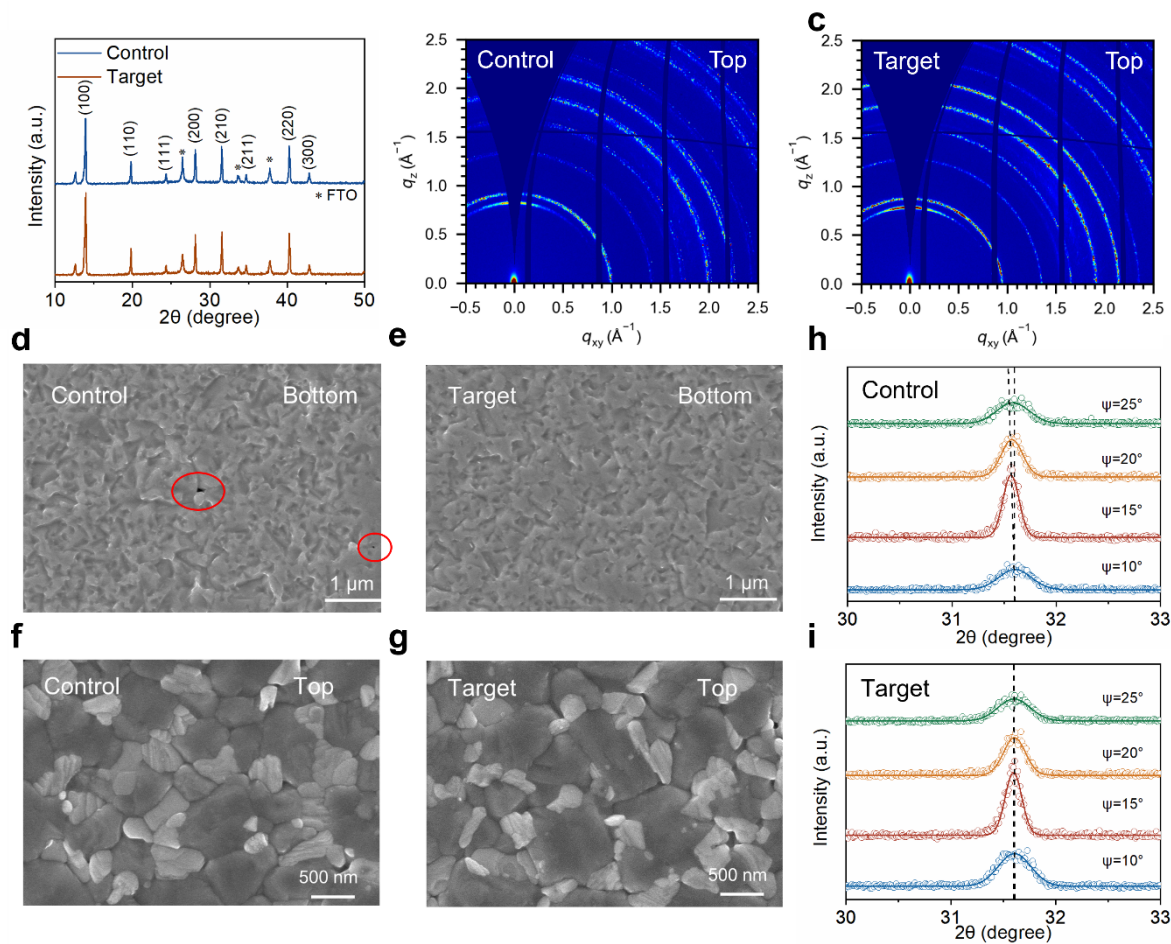
The SACH driven molecular ordering at the HTL layer plays a pivotal role in regulating perovskite crystallization and film morphology. By enhancing precursor-substrate interactions, the ordered SACH interface enables more uniform nucleation, thereby yielding higher-quality perovskite films at the buried interface. X-ray diffraction (XRD) patterns (Figure 2a) and depth-resolved grazing incidence wide-angle X-ray scattering (GIWAXS) (Figure 2b-c, Figure S4) show significantly intensified (100) diffraction peaks for perovskite films deposited on the SACH layer. These results underscore that the ordered SAM not only facilitates uniform nucleation but also drives preferential vertical crystallographic orientation.

Cross-sectional scanning electron microscopy (SEM) imaging provides further evidence of improved buried interface quality. As shown in Figure S5, perovskite films deposited on pristine Me-4PACz exhibit numerous voids at the perovskite/HTL interface, arising from

insufficient interfacial wetting and localized gaps caused by molecular aggregation. In sharp contrast, the NiO_x/SACH substrates yield densely connected buried interfaces with no visible voids (Figure 2d-e). These dense interfaces are attributed to the molecular-bridging effects provided by CE, which enhance interfacial adhesion and continuity^[22]. Top-view SEM images (Figure 2f-g) further reveal that perovskite films on the SACH layer possess larger grains, fewer grain boundaries, and narrower grain-size distributions, all of which contribute to reduced trap densities and minimized grain-boundary recombination.

Beyond morphological regulation, the SACH-engineered interface also facilitates effective stress relaxation within the perovskite lattice. Depth-dependent GIXRD measurements (Figure 2h-i) indicate that the SACH-modified perovskite films experience reduced tensile strain. While the control films' diffraction peaks shift to lower angles as the incidence angle rises—suggesting residual tensile stress—those grown on SACH layer exhibit negligible peak shifts, signifying the elimination of inhomogeneous structural distortion across the film depth. Such effective stress release is expected to benefit both charge transport and long-term operational stability^[23].

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Figure 2. The impacts of SACH layer on perovskite films crystallization and morphology. (a) XRD patterns, (b, c) GIWAXS images, SEM images of (d, e) the bottom surface, (f, g) the top surface and (h, i) GIXRD profiles of perovskite films grown on NiO_x/Me-4PACz and NiO_x/Me-4PACz+CE.

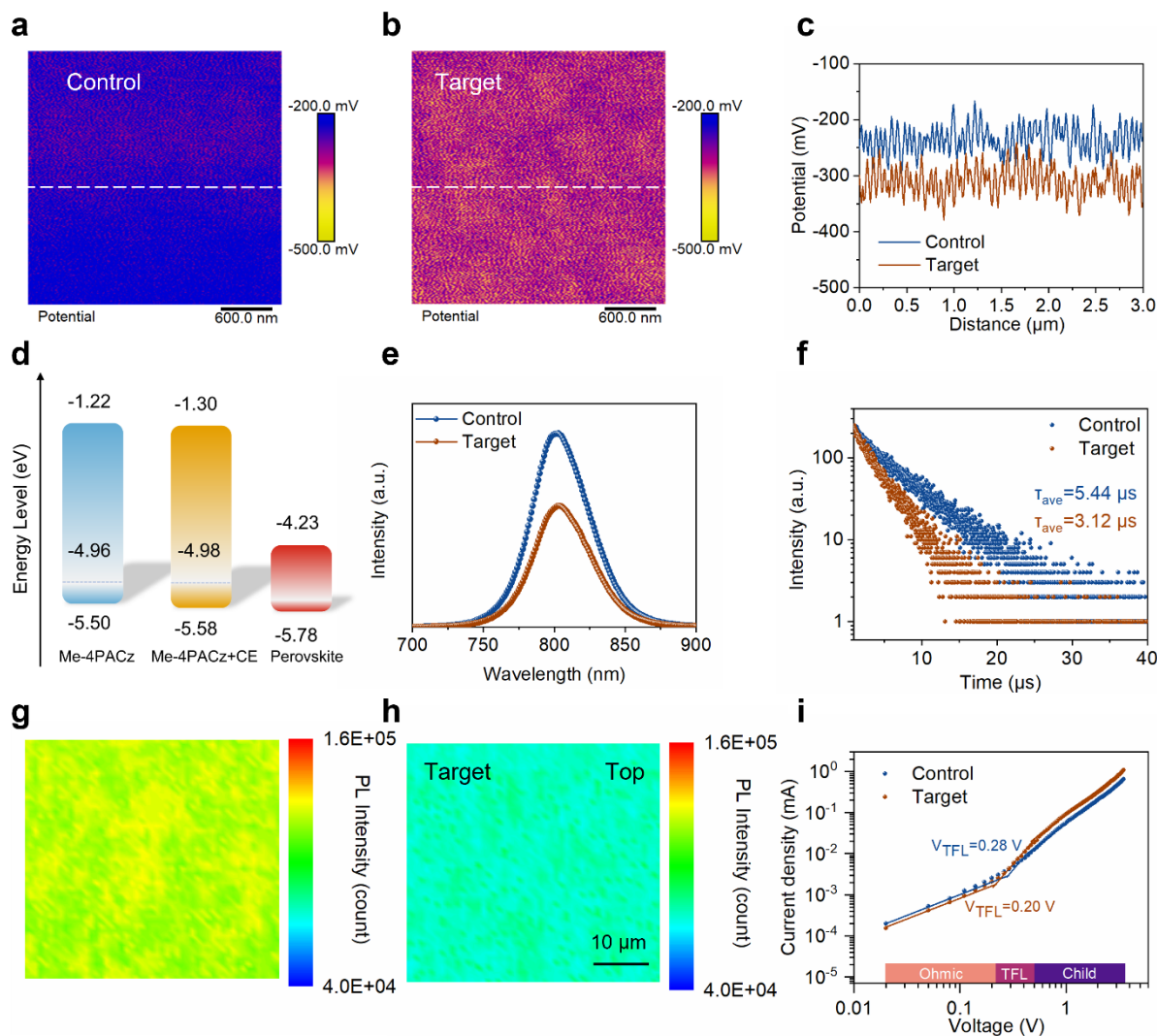
Beyond enhancing the perovskite quality, the modification of FTO/NiO_x with the SACH layer also substantially optimizes the electronic landscape at the buried perovskite interface. Kelvin probe force microscopy (KPFM) images reveal a lower contact potential difference (CPD) for the NiO_x/SACH layer film compared to the NiO_x/Me-4PACz film (Figure 3a-c), consistent with an increased interfacial work function. Ultraviolet photoelectron spectroscopy (UPS) measurements further confirm that the SACH layer increases the work function from 4.96 eV to 4.98 eV, attributed to its enhanced molecular dipole^[24] (Figure S6). Correspondingly, CE incorporation downshifts the valence band minimum (VBM) from -5.50 eV to -5.58 eV (Figure 3d), yielding improved energetic alignment with the perovskite absorber and thereby facilitating hole extraction and transport^[25].

The improved interfacial energetics are directly reflected in carrier recombination dynamics. Steady-state photoluminescence (PL) spectra show similar peak positions around 800 nm for both perovskite films, while the perovskite on the SACH layer exhibits markedly reduced PL intensity (Figure 3e), implying that the SACH layer extracts photogenerated holes more efficiently than the pristine Me-4PACz, surpassing the radiative recombination process^[26-27]. Time-resolved PL measurements (TRPL) further reveal a pronounced reduction in carrier lifetime from 5.44 μs to 3.12 μs for SACH-based films (Figure 3f and Table S2), confirming accelerated hole extraction driven by the optimized interfacial energy landscape^[28-29]. PL mapping (Figure 3g-h) further reveals that the perovskite film based on the SACH layer exhibits more uniform emission with overall lower PL intensity under top side excitation, which points to improved hole collection efficiency that suppresses radiative recombination at the interface. Meanwhile, buried-interface PL mapping shows uniformly higher emission intensity, indicating reduced non-radiative recombination due to effective passivation of interfacial defects by the SACH layer (Figure S7). The defect passivation mechanism is attributed to the specific chemical interactions introduced by CE. The hydroxyl (-OH) group of CE likely forms O-H...I hydrogen bonding interactions with iodide ions. This strong hydrogen-bonding interaction effectively anchors the surface iodide ions, thereby suppressing the generation of iodine vacancies and mitigating ion migration at the buried interface^[30-31].

These interfacial improvements are quantitatively reflected in space charge limited current (SCLC) measurements. As shown in Figure 3i, the trap-filling limit voltage (V_{TFL}), defined as

the intersection of the ohmic region and the trap-filling region, drops from 0.28 V to 0.20 V upon incorporating CE, corresponding to a reduced trap density from $1.21 \times 10^{15} \text{ cm}^{-3}$ to $8.63 \times 10^{14} \text{ cm}^{-3}$ (Table S3), demonstrating that the SACH layer could minimize non-radiative recombination and promote higher hole mobility, collectively leading to superior device performance^[32].

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Figure 3. SACH layer modulates energy band alignment, crystallinity, and carrier dynamics. KPFM characterization of (a) NiO_x/Me-4PACz and (b) NiO_x/Me-4PACz+CE. (c) Contact potential different (CPD) variations between NiO_x/Me-4PACz and NiO_x/Me-4PACz+CE films. (d) Energy alignment showing the VBM and conduction band minimum (CBM). (e) PL spectra and (f) TRPL decay curves of perovskite layers on NiO_x/Me-4PACz and NiO_x/Me-4PACz+CE. The PL mapping images of perovskite films deposited on (g) NiO_x/Me-4PACz and (h) NiO_x/Me-4PACz+CE. (i) SCLC analysis of the hole-only devices for hole mobility extraction.

To evaluate the effect of SACH strategy on the photovoltaic performance and long-term stability of inverted PSCs, we fabricated PSCs with a structure of FTO/NiO_x/Me-

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1 4PACz+CE/Cs_{0.05}MA_{0.1}FA_{0.85}PbI₃/C₆₀/BCP/Cu (Figure 4a). Statistical analysis of PV
 2 parameters confirmed high reproducibility with an average PCE of 26.84% based on SACH
 3 layer devices, which was markedly greater than that of the control device (25.33%) (Figure 4b).
 4 The champion device achieved a PCE of 27.06%, V_{oc} of 1.195 V, J_{sc} of 26.17 mA cm⁻², and FF
 5 of 86.56%, outperforming the control device (PCE of 25.66%, V_{oc} of 1.164 V, FF of 84.63%)
 6 (Figure 4c). Notably, a certified efficiency of 26.86% was independently verified for the SACH-
 7 based device (Figure S9), representing the highest certified PCE among recent regular-bandgap
 8 inverted PSCs employing Me-4PACz-based interfacial engineering (Table S5). Additionally,
 9 SACH-based devices exhibit markedly reduced hysteresis relative to the control (Figure S10
 10 and Table S4). The stabilized powers output (SPOs) at the maximum power point tracking
 11 confirms a stabilized PCE of 26.71% (Figure S11), consistent with J–V measurements. The
 12 integrated J_{sc} values from external quantum efficiency (EQE) spectra of the target and control
 13 cells are 25.28 and 25.01 mA cm⁻², respectively (Figure S12), aligning with the *J*-*V*
 14 measurements. Importantly, the generality of the SACH strategy is further validated by its
 15 successful extension to other widely used carbazole-based SAMs, specifically [2-(9H-carbazol-
 16 9-yl)ethyl]phosphonic acid (2PACz), [2-(3,6-dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic
 17 acid (MeO-2PACz), and [4-(3,6-diphenyl-9H-carbazol-9-yl)butyl]phosphonic acid (Ph-4PACz)
 18 (Figure S13), indicating that the SACH concept is broadly applicable to carbazole-based
 19 phosphonic acid SAMs with varying substituents.

20 Device physics analysis reveals that the performance gains originate from suppressed
 21 recombination, favorable energy-level alignment and improved charge transport^[33–35]. The
 22 reduced ideality factor (*n*) from 1.41 (control) to 1.17 (SACH-modified device) indicates
 23 suppressed trap-assisted recombination at the buried interface^[36] (Figure 4d). Consistently, the
 24 calculated maximum fill factor (FF_{max}) increases from 86.40% to 88.40% (Figure S14),
 25 indicating mitigation of non-radiative losses^[37]. The corresponding decrease in FF loss
 26 associated with charge transport—from 2.19% to 1.90%—signifies improved interfacial charge
 27 transport at the HTL/perovskite interface^[19, 22]. Dark J–V measurements further reveal reduced
 28 leakage current and steeper diode characteristics for SACH-modified devices (Figure 4e).
 29 Electrochemical impedance spectroscopy (EIS) further confirms that the SACH-modified
 30 device possesses decreased series resistance and increased recombination resistance (Figure
 31 4f), indicative of faster charge transport and suppressed carrier recombination^[38, 39]. Mott-
 32 Schottky analysis reveals a rise in built-in potential (0.97 V vs. 0.92 V for the control) (Figure
 33 4g), facilitating improved charge separation and higher V_{oc} ^[40]. These results collectively
 34 indicate that the efficiency enhancement mainly originates from suppressed non-radiative
 35 recombination at the buried interface rather than changes in light absorption.

Long-term stability testing of the unencapsulated devices demonstrates the robustness of the SACH interface. For unencapsulated devices, the SACH-based PSCs retain nearly 100% of their initial efficiency after 1368 h of continuous storage, whereas the control devices degrade to 92.20% of their initial performance (Figure S15), indicating intrinsic interfacial stability improvement. Furthermore, operational stability was evaluated via maximum power point tracking (MPPT) on encapsulated devices in ambient air under continuous 1-sun illumination at 65 °C. The SACH-modified device maintains 88.6% of its initial efficiency after 500 h of continuous operation. In contrast, the control devices retain only 65.4% of their initial efficiency after 380 h (Figure 4h), further confirming that the SACH interface effectively suppresses operational degradation.

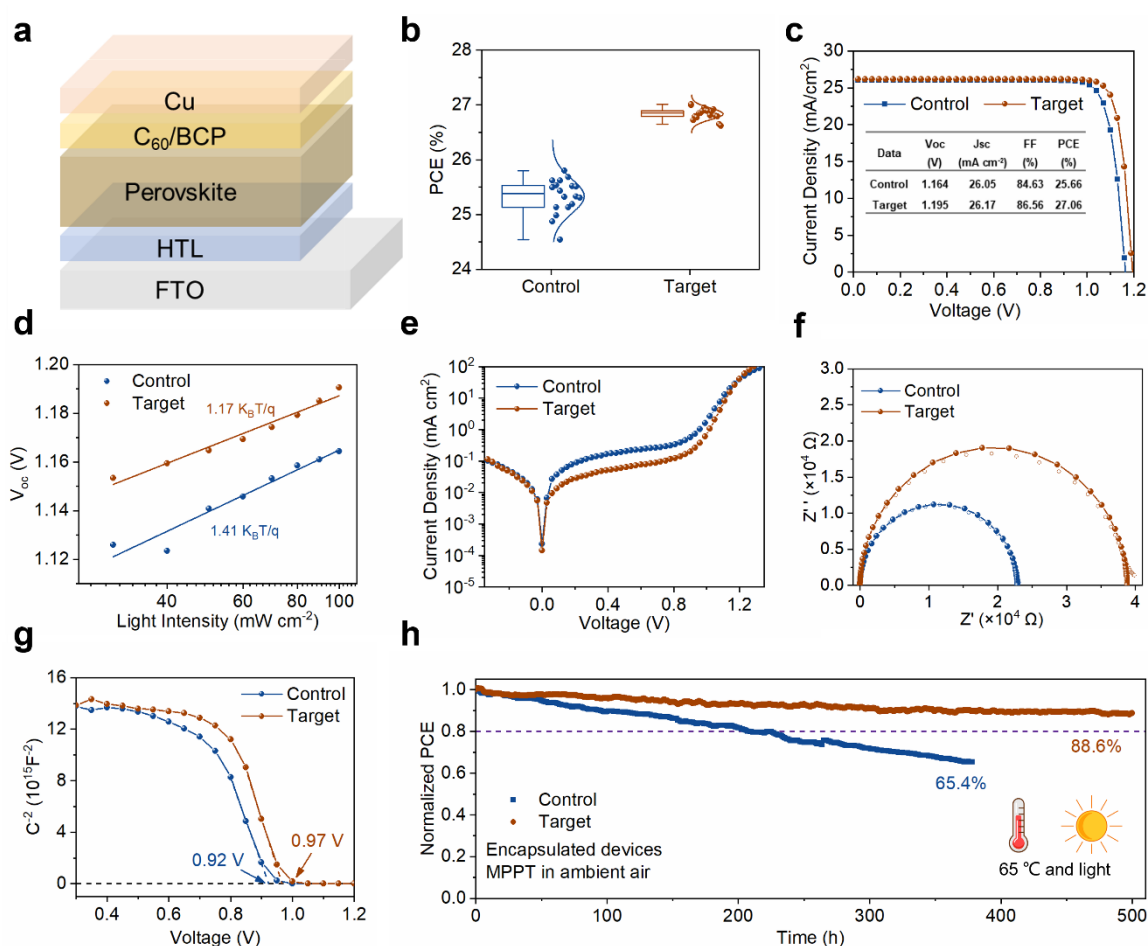
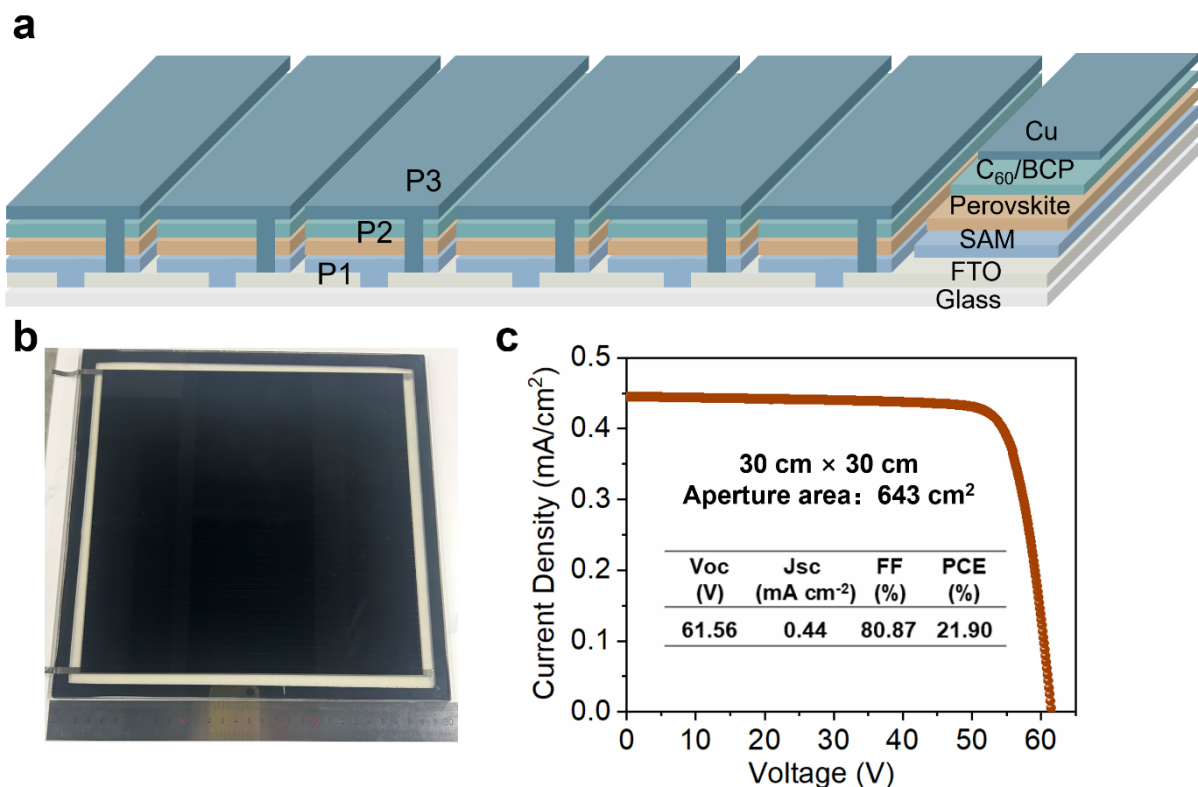


Figure 4. Impact of SACH layer on device performance. (a) Schematic illustration of the device configuration. (b) Statistics distribution of the PCE. (c) Optimized $J-V$ plots for the best-performing devices (AM 1.5G), (d) Light intensity dependence of V_{oc} , (e) Dark $J-V$ curves, (f) Nyquist spectra, (g) Mott-Schottky analysis and (h) Normalized PCEs of encapsulated devices based on $\text{NiO}_x/\text{Me-4PACz}$ and $\text{NiO}_x/\text{Me-4PACz}+\text{CE}$ measured under continuous maximum power point tracking (MPPT) at 65 °C in ambient air.

To further assess the potential of the SACH-based method for commercial sub-module fabrication, a scalable processing strategy for large-area modules was introduced as illustrated in Figure 5a. Figure 5b displays a 30 × 30 cm² perovskite solar module (PSM), the SACH interfacial layer significantly improves the wettability of the perovskite precursor on the substrate, ensuring the deposition of homogeneous, pinhole-free films over a large area. Consequently, an impressive efficiency of 21.90% was achieved on an aperture area of 643 cm² (Figure 5c), demonstrating the strong potential of the SACH strategy for scalable and commercially relevant perovskite photovoltaic manufacturing.

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Figure 5. Fabrication of scalable large-area modules. (a) Schematic illustration of the scalable fabrication for large-area PSMs, including the associated laser patterning steps. (b) Digital image of a 30 cm × 30 cm module. (c) *J*-*V* characteristics of the best module.

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3. Conclusion

In summary, we have established a synergistically assembled chemically heterogeneous (SACH) interfacial layer by integrating Me-4PACz with CE on the NiO_x surface, providing an effective route to engineer the buried interface of inverted perovskite solar cells. Through cooperative molecular assembly, the carbazole moiety of CE engages in π - π interactions with Me-4PACz, suppressing its intrinsic self-aggregation and promoting the formation of a densely

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1packed yet chemically heterogeneous monolayer. Simultaneously, the hydroxyl group of CE
2enhances interfacial wettability and establishes O–H···I⁻ interactions with the perovskite, which
3effectively facilitate crystal nucleation, improve film quality, and passivate interfacial defects.
4Benefiting from these synergistic effects, devices incorporating the SACH layer achieved a
5substantially enhanced PCE of 27.06% (certified efficiency of 26.86%), surpassing the 25.66%
6obtained from the Me-4PACz-only counterpart, along with notable improvements in stability.
7Under MPPT operation in ambient air at 65 °C, encapsulated SACH-based devices retained
888.6% of their initial efficiency after 500 h, confirming their enhanced operational stability.
9Furthermore, large-area perovskite modules (30 × 30 cm²) fabricated using the SACH strategy
10exhibited a high efficiency of 21.90%, demonstrating its strong potential for scalable and
11commercially relevant perovskite photovoltaic manufacturing. These results demonstrate that
12cooperative molecular assembly offers a generalizable and industry-compatible approach for
13creating structurally ordered and electronically favorable interfaces, establishing a
14generalizable strategy toward high-efficiency, stable inverted PSCs and scalable perovskite
15photovoltaic modules.

16

17**Conflict of Interest**

18 The authors declare no conflict of interest.

19

20**Acknowledgements**

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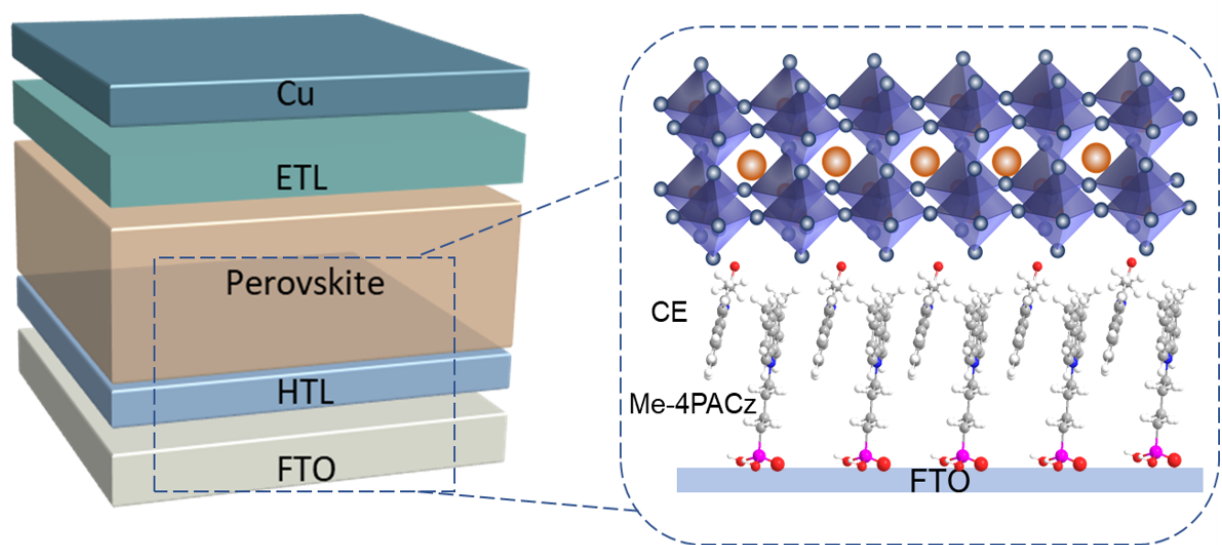
12

1 A synergistically assembled chemically heterogeneous interface, integrating Me-4PACz
2 with carbazole-9-ethanol, is engineered to transform inverted perovskite solar cells.
3 Cooperative π - π interactions suppress molecular aggregation and improve interfacial order,
4 while hydroxyl-mediated passivation enhances perovskite crystallization. This interfacial
5 design simultaneously boosts efficiency, stability, and scalability, delivering a power
6 conversion efficiency of 27.06% (certified 26.86%) and a module efficiency of 21.90%.

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9 Y.B. Lou*, T.Y. Zhang*

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11 **Cooperative Assembly Drives Buried Interface Engineering for Efficient Inverted**
12 **Perovskite Solar Cells**

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14

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Supporting Information

Cooperative Assembly Drives Buried Interface Engineering for Efficient Inverted Perovskite Solar Cells

Ranran Xu, Ni Zhang, Shengnan Wang, Fengqiao Xu, Weidong Lin, Ruoting Liu, Lu Pan, Yixin Zhao, Zhixiao Qin*, Yongbing Lou*, Taiyang Zhang*

Experimental Section

1. Materials

NiO_x nanoparticles (99.999%) were sourced from Suzhou Liwei Technology. [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz, >98%) was purchased from TCI, and carbazole-9-ethanol (CE, >98%) was purchased from Tansoole. 2,2',7,7'-tetrakis(N,N-di-p-methoxyphenylamine)-9,9'-spirobifluorene (Spiro-OMeTAD, 99.9%), 4-tert-butylpyridine (TBP, 96%) and lithium bis(trifluoromethanesulfonyl)imide (LiTFSI, 99%), formamidinium iodide (FAI, 99.9%), methylammonium chloride (MACl, 99.9%), lead iodide (PbI₂, 99.99%), methylammonium iodide (MAI, 99.9%), were purchased from Xi'an Yuri Solar Co., Ltd. Cesium iodide (CsI, 99.999%) was purchased from Aladdin Biochemical Technology Co., Ltd.). Bathocuproine (BCP, 99.5%) was purchased from Zhoushan Huazhou Chemical Co., Ltd. and C₆₀ (99.5%) was purchased from Borun New Material Technology Co., Ltd. N,N-dimethylformamide (DMF, 99.8%), dimethyl sulfoxide (DMSO, 99.7%), ethanol (99.5%), isopropanol (IPA, 99.5%), chlorobenzene (CB, 99.8%) and ethanol absolute (EtOH, superdry) were purchased from J&K Scientific LLC. FTO conductive glasses (FTO, 15 Ω sq⁻¹) were purchased from Suzhou Shangyang Solar Technology Co., Ltd. All materials were used directly as received without further purification.

2. Perovskite precursor and film preparation

The perovskite precursor (Cs_{0.05}MA_{0.10}FA_{0.85}PbI₃, 1.5 M) was prepared by mixing stoichiometric amounts of CsI, MAI, FAI and PbI₂ in DMF: DMSO (4:1, v/v). The precursor was further supplemented with 5 mol% excess PbI₂ and 12.5 mol% MACl to facilitate crystallization and phase conversion. Following 1 h of sonication and filtration (0.22 μm PTFE),

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60 μL of the precursor was deposited via spin-coating at 1000 rpm (10 s) and 5000 rpm (30 s), utilizing acceleration rates of 1000 and 3000 rpm s^{-1} . 200 μL of CB was rapidly added 5 s before the spinning ended. The film was then immediately annealed at 100 $^{\circ}\text{C}$ for 30 min.

3. Small-area devices fabrication

Fluorine-doped SnO_2 (FTO) glass substrates were sequentially washed using detergent, deionized water and ethanol, with residual liquids blown off using nitrogen gas. The cleaned FTO was exposed to ultraviolet ozone treatment for 15 min prior to use. NiO_x nanoparticles (10 mg mL^{-1} in deionized water) were deposited by spin-coating at 3000 rpm for 30 seconds, and annealed in air at 120 $^{\circ}\text{C}$ for 10 min, then immediately moved to a nitrogen glovebox. Subsequently, Me-4PACz solution (0.66 mg mL^{-1} in ethanol) was spin-coated on NiO_x layer at 3000 rpm for 30 s, followed by annealing at 100 $^{\circ}\text{C}$ for 10 min. For the target device, a mixed solution of Me-4PACz (0.66 mg mL^{-1}) and CE (0.51 mg mL^{-1}) was deposited onto the NiO_x layer under the same conditions, followed by the same annealing process. Perovskite films were subsequently deposited on HTLs according to the previously stated method. A passivation layer consisting of 1.5 mg mL^{-1} PDAI₂ (dissolved in IPA) was spin-coated at 5000 rpm for 30 s and annealed at 100 $^{\circ}\text{C}$ for 5 min. The device was completed by thermally evaporating C_{60} (20 nm), BCP (6 nm), and Cu (80 nm) under vacuum of 5×10^{-5} Torr with an aperture area of 0.062 cm^2 .

4. Large-area devices fabrication

The series interconnection of the module was defined through laser scribing of P1, P2, and P3 lines. The respective line widths were 13 μm for P1, 54 μm for P2, and 18 μm for P3. The spacing between P1 and P2 was set at 40 μm , and 30 μm between P2 and P3. A NiO_x film was subsequently deposited by radio-frequency (RF) magnetron sputter. Next, a mixed solution of Me-4PACz and CE was slot-die coated onto the FTO/ NiO_x substrate. For the perovskite layer, a 1.4 M precursor solution was slot-die coated at a speed of 20 mm s^{-1} , with a solution supply rate of 20 $\mu\text{L s}^{-1}$, and a substrate-to-die gap of 110 μm , followed by a vacuum-flashing step and annealing at 150 $^{\circ}\text{C}$ for 10 min. Finally, C_{60} , BCP and Cu were thermally evaporated following the same processes as those employed for the small-area solar cells.

5. DFT

Quantum chemical computations were performed via the Gaussian09 software^[1]. Initial geometries were optimized employing the B3LYP density functional with 6-311G (d, p) basis set. To verify that the optimized geometries represented true local minima on the potential

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energy surface, frequency analyses were performed at the same theoretical level. Subsequently, the interaction energies were calculated using the projector augmented-wave (PAW) method within the VASP code^[2]. Exchange-correlation potentials were treated using the Perdew–Burke–Ernzerhof (PBE) generalized gradient approximation (GGA)^[3]. Long-range van der Waals interactions were included using the DFT-D3 dispersion correction method^[4]. Calculations employed a 400 eV cutoff energy, requiring energy convergence to 1×10^{-5} eV/atom and the force relaxation below 0.01 eV/Å. A $1 \times 1 \times 1$ k-point was used for Brillouin zone sampling.

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10 6. Measurements and Instruments

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12 The scanning electron microscope (SEM) images were acquired using FESEM (SU8010, 13 Hitachi, Japan). Atomic force microscope (AFM) and Kelvin probe force microscopy (KPFM) 14 measurements were obtained with Bruker Dimension® ICON. Ultraviolet photoelectron 15 spectroscopy (UPS) spectra were recorded on ESCALAB 250Xi equipped with an Al K α X- 16 ray source and a He I α photon source (21.22 eV), respectively. The Fourier transform infrared 17 (FTIR) spectroscopy tests were performed on Thermo Scientific Nicolet iS20 (USA). The 18 Grazing-incidence X-ray diffraction (GIXRD) pattern were collected at grazing incident angles 19 of 10°, 15°, 20° and 25° using an X-ray diffraction (XRD, DX-2800 X, Haoyuan, Liaoning, 20 China) to analyze the phases of the samples and films. An ultraviolet and visible spectrum (UV- 21 vis, Cary60, Agilent, Palo Alto, CA, USA) was employed to study the absorption intensity of 22 the thin films. Grazing-incidence wide-angle X-ray scattering (GIWAXS) measurements were 23 performed at BL14B1 line station of Shanghai Synchrotron Radiation Facility, utilizing a beam 24 wavelength of 0.12398 nm, and a grazing-incidence angle of 0.3°. Photoluminescence (PL) 25 spectra and time-resolved photoluminescence (TRPL) spectra were measured by Edinburgh 26 Instruments FLS1000.

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28 7. Device Characterizations

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30 A Keithley 2401 source meter coupled with an Enlitech 3A AM 1.5G simulator (Shanghai, 31 China) was employed to measure current density-voltage (J-V) curves, following intensity 32 calibration with a silicon reference. Scans were performed in both forward (from -0.1 to 1.3 V) 33 and reverse (from 1.3 to -0.1 V) directions. A voltage step of 30 mV was applied with a delay 34 of 50 ms between points. The external quantum efficiency (EQE) curves were measured by an 35 Enlitech QE-3011 system. The electrochemical impedance spectroscopy (EIS) and Mott- 36 Schottky measurements were conducted on a CHI660D electrochemical workstation (CHI

Instrument, Shanghai, China). The space charge limit current (SCLC) test and dark current test were performed on a Keithley 2401 source meter. For the stability test, unencapsulated devices underwent aging tests in nitrogen atmosphere.

Supplementary Note

1. The space charge limit current (SCLC)

The hole-only devices for the SCLC measurements were fabricated with the structure of FTO/NiO_x/SAMs/perovskite/Spiro-OMeTAD/Cu. The dark I-V curves were recorded using a Keithley 2401 Source Meter. The trap-state density of perovskite films can be calculated as follows equation^[5]:

$$N_{trap} = \frac{2V_{TFL}\epsilon_r\epsilon_0}{qL^2} \quad (1)$$

In this equation, N_{trap} represents the trap density, while V_{TFL} and L denote the trap-filled limit voltage and perovskite film thickness (800 nm), respectively. Constants ϵ_r , ϵ_0 and q correspond to the relative dielectric constant (assumed to be 25 for FAPbI₃), vacuum permittivity, and elementary charge, respectively.

2. TRPL decay curve fitting

The decay curve fitting of TRPL follows the bi-exponential equation:

$$y = y_0 + A_1e^{-x/\tau_1} + A_2e^{-x/\tau_2} \quad (2)$$

The average lifetime can be calculated as:

$$\tau_{avg} = \frac{(A_1\tau_1^2 + A_2\tau_2^2)}{A_1\tau_1 + A_2\tau_2} \quad (3)$$

where A_1 and A_2 are the correlation coefficient of carrier lifetime τ_1 and τ_2 , respectively.

3. Hysteresis factor

The following expression was employed to calculate the hysteresis index (HI):

$$HI = \frac{PCE_{reverse} - PCE_{forward}}{PCE_{reverse}} \quad (4)$$

4. Light intensity dependence of V_{oc}

The carrier recombination kinetics were elucidated by analyzing the dependence of the device V_{oc} on the incident light intensity following:

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$$V_{oc} = \frac{nkT \ln L}{q}$$

(5)

where n , k , T and L represent the ideality factor, Boltzmann constant, temperature, and light intensity, respectively, with q denoting the elementary charge.

5. Fill factor (FF) loss analysis

FF losses in high-efficiency PSCs primarily stem from non-radiative recombination and charge transport limitations. The theoretical maximum FF (FF_{max}) is empirically estimated via:

$$FF_{max} = \frac{v_{oc} - \ln(v_{oc} + 0.72)}{v_{oc} + 1}$$

(6)

where $v_{oc} = \frac{V_{oc}}{nkT/q}$, Ideality factors were derived from the semi-logarithmic plot of V_{oc} versus light intensity, as presented in Figure S18 (Supporting Information).

6. Mott-Schottky plots

The capacitance-voltage (C - V) measurements in the Mott-Schottky configuration were performed to determine the interfacial carrier density. The corresponding relation is:

$$\frac{1}{C^2} = \frac{2(V_{bi} - V)}{A^2 q \epsilon_0 \epsilon N_A} \quad (7)$$

where V_{bi} , V and A represent the built-in potential, applied bias, and active area, respectively. N_A is the doping concentration, q is the elementary charge, ϵ is the dielectric constant, and ϵ_0 is the vacuum permittivity.

Supplementary Figures and Tables

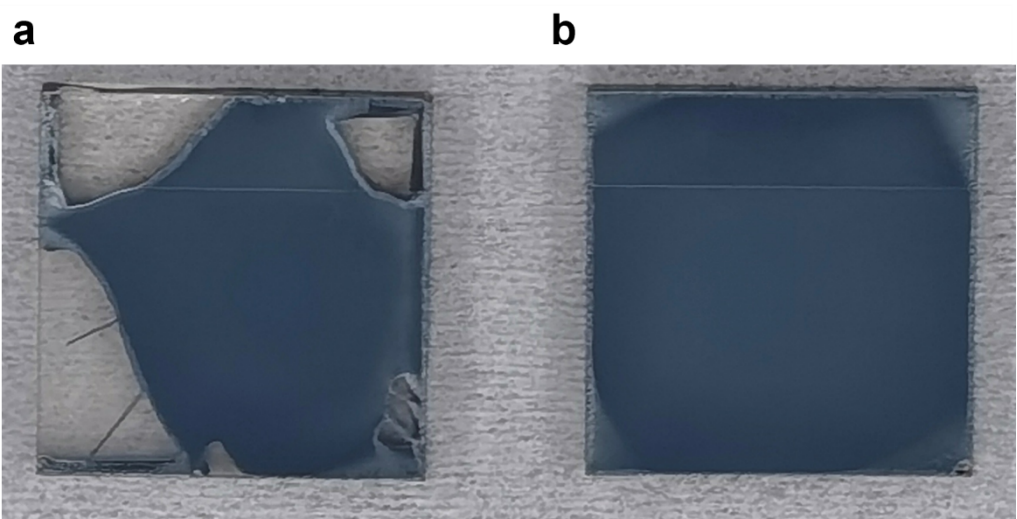


Figure S1. The image of perovskite films deposited on (a) FTO/NiO_x/Me-4PACz and (b) FTO/NiO_x/Me-4PACz+CE.

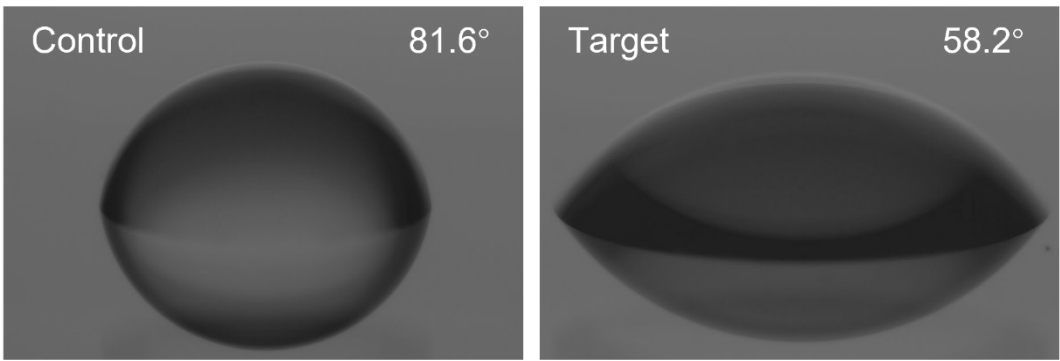


Figure S2. Water contact angle on (a) FTO/NiO_x/Me-4PACz and (b) FTO/NiO_x/Me-4PACz+CE.

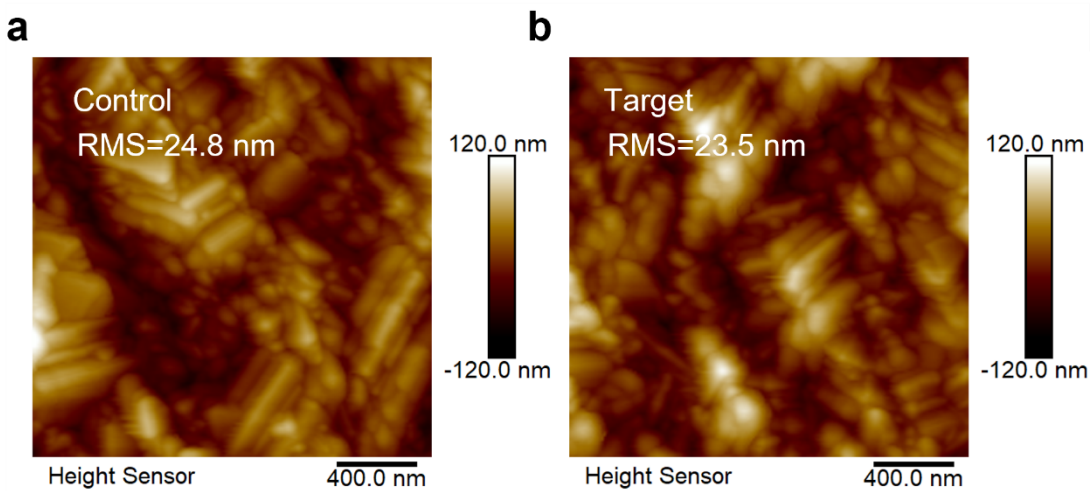


Figure S3. AFM height profiles of (a) Me-4PACz and (b) Me-4PACz+CE on FTO/NiO_x.

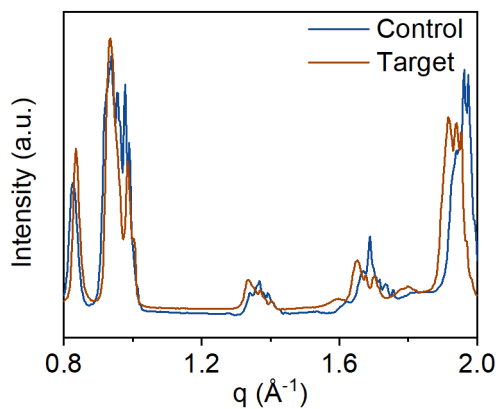
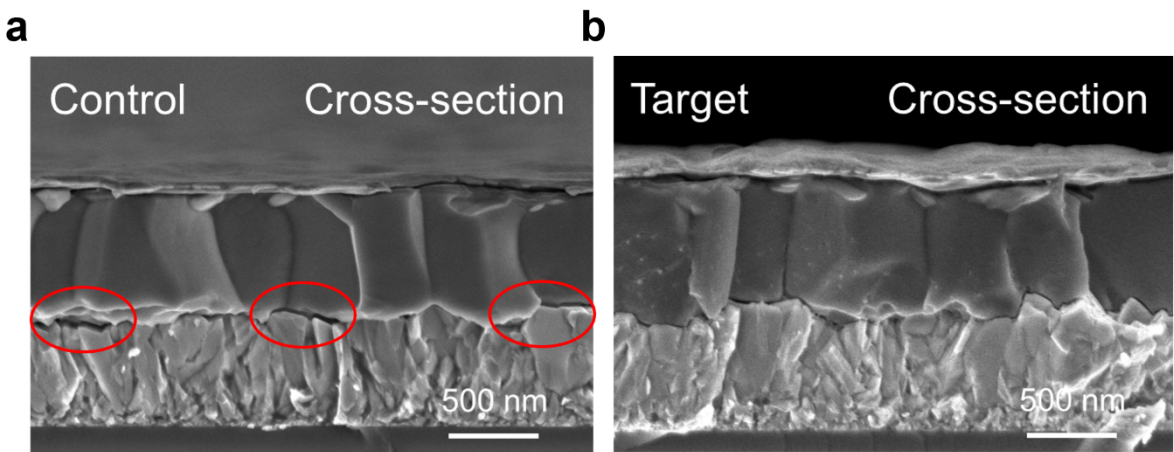
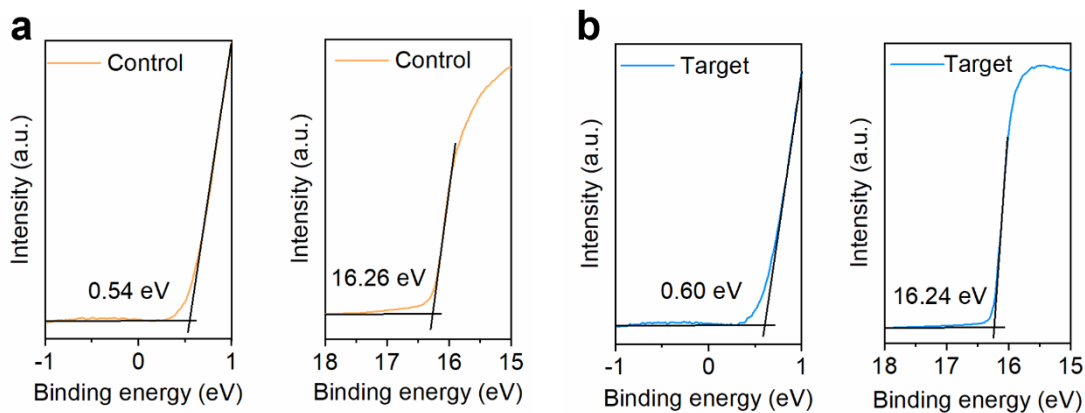


Figure S4. Azimuthally integrated GIWAXS profiles of perovskite films deposited on $\text{NiO}_x/\text{Me-4PACz}$ and $\text{NiO}_x/\text{Me-4PACz+CE}$ coated FTO.



1
2 **Figure S5.** The cross-section SEM images of PSCs deposited on (a) NiO_x/Me-4PACz and (b)
3 NiO_x/Me-4PACz+CE.



6
7 **Figure S6.** UPS spectra of (a) NiO_x/Me-4PACz and (b) NiO_x/Me-4PACz+CE films.

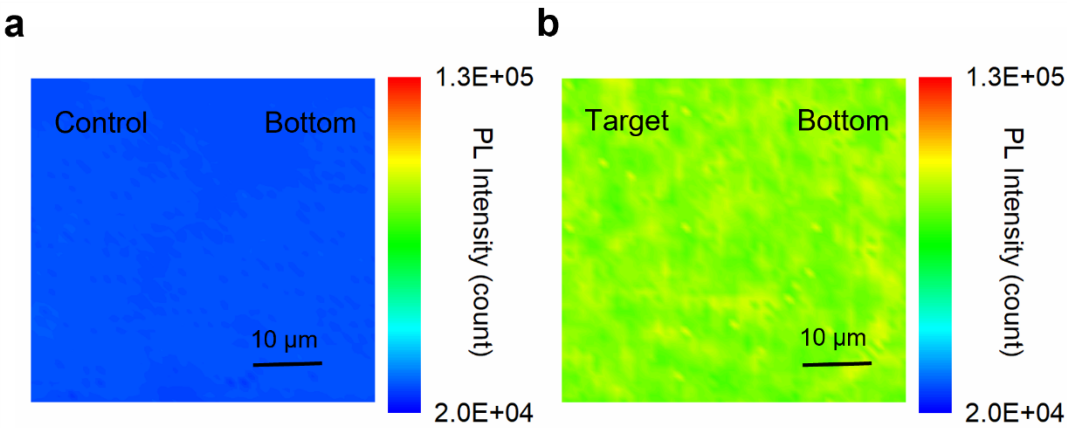


Figure S7. PL mapping images of bottom surface of the perovskite films deposited on (a) $\text{NiO}_x/\text{Me-4PACz}$ and (b) $\text{NiO}_x/\text{Me-4PACz+CE}$.

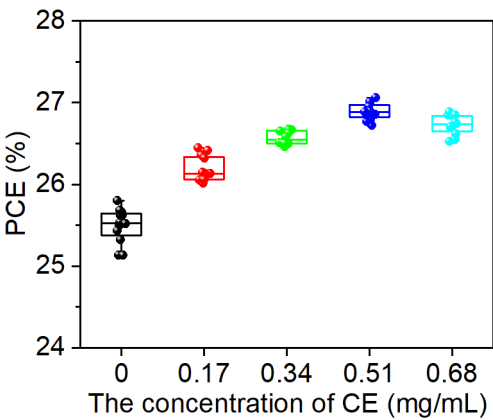


Figure S8. Statistical distribution of the power conversion efficiencies (PCEs) based on the Me-4PACz SAMs modified with different concentrations of CE.

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天津市计量监督检测科学研究院
Tianjin Institute of Metrological Supervision and Testing
检测报告
Test Report

报告编号: JLY20251385W
Report No.

共 4 页 第 1 页
Page No. 4-1

样品名称 Sample Name	钙钛矿太阳能电池				
规格型号 Specification Type	Pvsktech-PSCs	商标/出厂编号 Brand/Serial Number	/ #202512-5-23NJ	生产日期/批号 Produce Date/ Serial Number	/
样品等级 Sample Grade	/	样品描述 Sample Description	外观良好, 无 明显可见缺陷	样品接收时间 Sample receive time	2025-12-24
委托日期 Delivery Date	2025-12-24	送样人员 Delivered by	/	样品数量 Sample Quantity	1块
委托单位名称 及联系电话 Name of Client	上海钙晶科技有限公司/15289484647				
生产单位名称 Manufacturer	上海钙晶科技有限公司				
检测时间 Test Time	2025-12-24	检测地点 Test Location	本院五号堤路院 区光伏产业计量 部106室	检测环境 Test Condition	温度: 25.1℃ 相对湿度: 36%
检测依据 Test Standard	IEC 60904-1:2020 《光伏器件 第1部分: 光伏电流-电压特性的测量》				
检测结论 Test Conclusion	1201040230084 天津市计量监督检测科学研究院 (检验检测专用章) (Stamp of Test Institution) 签发日期: 2026-01-08 Issue Date: Y M D				
备注 Remark	/				

监督
检测
040

编制: 周超
Compiled by

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批准: 柳云秀
Approved by

天津市计量监督检测科学研究院
Tianjin Institute of Metrological Supervision and Testing

检测报告
Test Report

报告编号: JLY20251385W 共 4 页 第 4 页
Report No. Page No.4-4

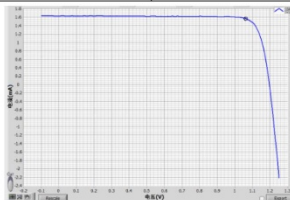
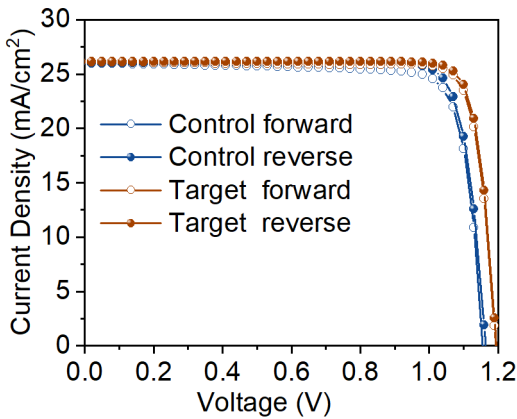
检 测 项 目 及 结 果								
检测项目 Test Item	光伏电流-电压特性 I-V Characteristic							
测试时间 Date	2025.12.24							
测试条件 Condition	使用稳态 AAA 级太阳模拟器, 在 AM1.5G, 1000W/m ² , 25.0°C条件下测试 Sample was tested under the condition of AM1.5G, 1000W/m ² , 25.0°C with a steady-state class calibrated AAA solar simulator							
有效面积 Active area	0.0620 cm ² 备注: 器件的有效面积是由带固定孔径的薄金属掩模板量化。 Remark: Designated area defined by thin metal aperture mask.							
样品编号 №	V _{OC} (V)	I _{SC} (mA)	J _{SC} (mA/cm ²)	P _{max} (mW)	V _{Fmax} (V)	I _{Fmax} (mA)	FF (%)	η (%)
20251385-1	1.196	1.623	26.17	1.665	1.060	1.571	85.78	26.86
测试程序 Test program settings	起始电压: Starting voltage:				+1.25 V			
	终止电压: Termination voltage:				-0.10 V			
	扫描间隔: Scan interval:				0.01 V			
	延迟时间: Delay time:				0.05 s			
I-V 曲线图 I-V curve								
备注 Remark	/							
—————以下空白————— Blank below								

Figure S9. Certificate report of the SACH-based device.

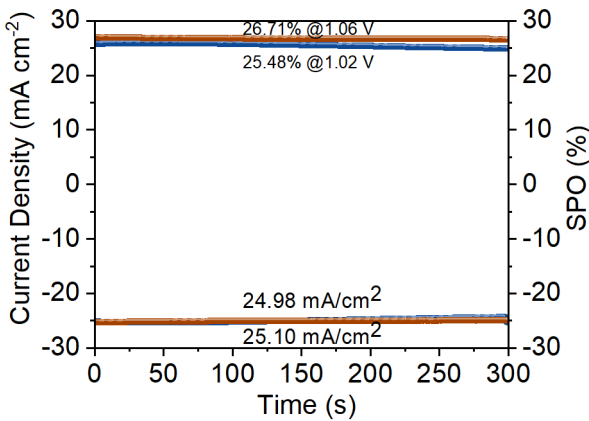


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2 **Figure S10.** Forward and reverse J - V scans of the inverted PSCs based on $\text{NiO}_x/\text{Me-4PACz}$
3 and $\text{NiO}_x/\text{Me-4PACz+CE}$.

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8 **Figure S11.** Steady-state power output of the devices based on $\text{NiO}_x/\text{Me-4PACz}$ and $\text{NiO}_x/\text{Me-}$
9 4PACz+CE .

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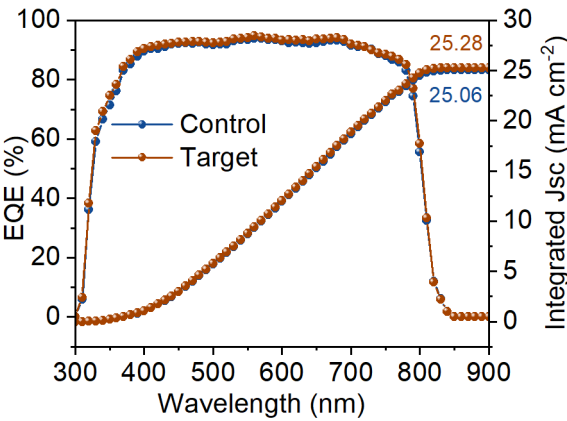


Figure S12. EQE spectra of the devices based on different HTLs.

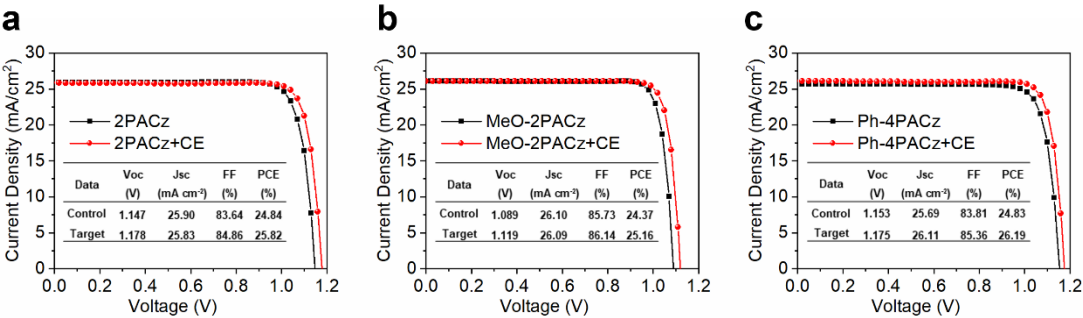


Figure S13. J-V curves and photovoltaic parameters of the devices based on different HTLs. (a) 2PACz and 2PACz+CE, (b) MeO-2PACz and MeO-2PACz+CE, (c) Ph-4PACz and Ph-4PACz+CE.

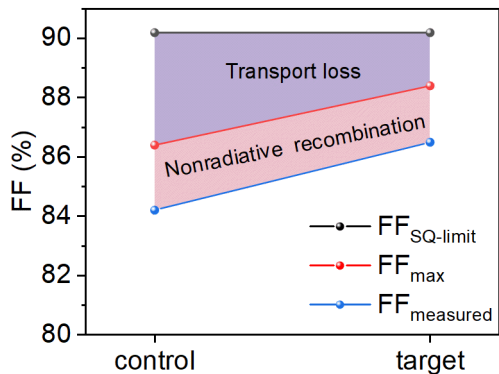


Figure S14. Detailed analysis of FF losses.

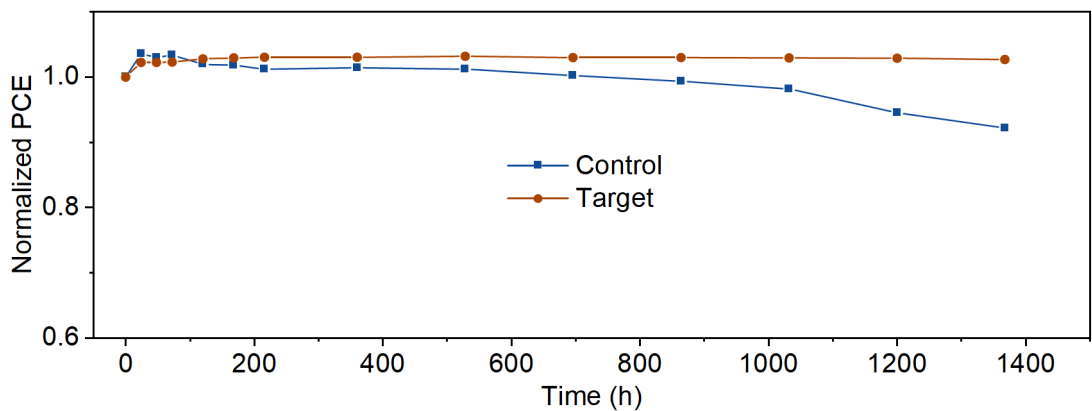


Figure S15. Normalized PCEs of unencapsulated devices based on $NiO_x/Me-4PACz$ and $NiO_x/Me-4PACz+CE$ stored under inert atmosphere.

1 **Supplementary Table S1.** UPS data of Me-4PACz and Me-4PACz+CE films.

Samples	E _g (eV)	WF (eV)	VB (eV)	CB (eV)
Me-4PACz	4.28	4.96	-5.50	-1.22
Me-4PACz+CE	4.28	4.98	-5.58	-1.30

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5**Supplementary Table S2.** The PL decay parameters for perovskite films deposited on
6NiO_x/Me-4PACz and NiO_x/Me-4PACz+CE interfaces.

Samples	A ₁	τ ₁ (μs)	A ₂	τ ₂ (μs)	τ _{ave} (μs)
Control	138.82	1.10	212.31	5.96	5.44
Target	159.97	0.95	231.69	3.52	3.12

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Supplementary Table S3. SCLC determined trap parameters for hole-only devices deposited on Me-4PACz and Me-4PACz+CE interfaces.

Samples	V_{TFL} (V)	N_t (10^{15} cm^{-3})
Control	0.28	1.21
Target	0.20	0.86

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Supplementary Table S4. Detailed J-V parameters under AM 1.5G illumination.

Devices	Scan	V_{oc} (V)	J_{sc} (mA cm^{-2})	FF (%)	PCE (%)	HI
Control	Forward	1.154	26.01	82.59	24.80	3.36%
	Reverse	1.164	26.05	84.63	25.66	
Target	Forward	1.193	26.11	85.47	26.11	1.58%
	Reverse	1.195	26.17	86.56	27.06	

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¹**Supplementary Table S5.** Summary of device performance parameters of regular-bandgap
²inverted PSCs based on various Me-4PACz interface engineering strategies from literature.

Modifier	V_{oc} (V)	J_{sc} (mA cm ⁻²)	FF (%)	PCE (%)	Certified PCE (%)	Ref.
MeO-2PACz	1.18	25.58	84.9	25.58	24.3	[6]
FBA	1.186	25.78	83.69	25.58	/	[7]
CzTPA	1.172	25.78	84.92	25.66	/	[8]
NA	1.201	26.30	84.50	26.69	26.54	[9]
Br-4PACz	1.187	25.79	85.82	26.28	25.80	[10]
ABT	1.199	25.57	84.27	25.75	25.45	[11]
DPPO-4PACz	1.181	26.07	83.41	25.67	/	[12]
TDCA	1.189	25.80	85.45	26.20	/	[13]
BPPA	1.19	25.86	85.31%	26.35	/	[14]
MCC	1.199	25.99	85.94	26.78	26.65	[15]
AG	1.184	26.06	86.65	26.74	26.21	[16]
4Br-BPA	1.182	26.24	85.74	26.59	26.12	[17]
N-DPPM	1.192	26.09	86.19	26.80	26.11	[18]
BzMIMBF ₄	1.171	26.20	86.17	26.45	26.37	[19]
MTBH	1.19	25.94	84.94	26.29	/	[20]
PPBCz-4	1.188	26.29	86.15	26.91	26.44	[21]
2PACz	1.185	26.6	85.8	27.1	26.4	[22]
CE	1.195	26.17	86.56	27.06	26.86	This work

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wavelength [nm]	EQE [%]
300	0
310	6.66
320	38.34
330	62.88
340	69.21
350	74.63
360	78.37
370	84.42
380	86.86
390	89.38
400	90.35
410	91.18
420	91.45
430	91.87
440	92.18
450	92.51
460	92.68
470	92.85
480	92.78
490	92.55
500	92.37
510	92.5
520	92.72
530	93.72
540	94
550	94.13
560	94.76
570	94.39
580	93.84
590	93.92
600	93.27
610	93.4
620	93.14
630	93.55
640	93.06
650	93.61
660	93.65
670	93.89
680	94.11
690	93.53
700	91.97

710	91.47
720	91.14
730	89.9
740	88.93
750	88.55
760	87.83
770	86.8
780	85.04
790	77.05
800	58.43
810	33.35
820	11.75
830	5.91
840	1.86
850	0
860	0
870	0
880	0
890	0
900	0

Reverse scan		Forward scan	
Voltage [V]	Current density [mA/cm ²]	Voltage [V]	Current density [mA/cm ²]
1.25	-36.01332	-0.13	26.1371
1.22	-14.54562	-0.1	26.13045
1.19	2.57141	-0.07	26.1227
1.16	14.31608	-0.04	26.11603
1.13	20.9169	-0.01	26.10882
1.1	24.01182	0.02	26.10295
1.07	25.2864	0.05	26.09473
1.04	25.78972	0.08	26.08662
1.01	25.99055	0.11	26.08058
0.98	26.07535	0.14	26.07223
0.95	26.11483	0.17	26.06422
0.92	26.13398	0.2	26.05633
0.89	26.14652	0.23	26.04807
0.86	26.15307	0.26	26.03957
0.83	26.15515	0.29	26.02982
0.8	26.1554	0.32	26.0205
0.77	26.15347	0.35	26.01215
0.74	26.15155	0.38	26.00202
0.71	26.14908	0.41	25.99192
0.68	26.14832	0.44	25.98395
0.65	26.14748	0.47	25.97548
0.62	26.14672	0.5	25.96445
0.59	26.14675	0.53	25.95308
0.56	26.14787	0.56	25.94147
0.53	26.14787	0.59	25.92827
0.5	26.14817	0.62	25.91485
0.47	26.15035	0.65	25.90305
0.44	26.15272	0.68	25.88892
0.41	26.15307	0.71	25.87663
0.38	26.1553	0.74	25.86628
0.35	26.15585	0.77	25.86155
0.32	26.157	0.8	25.8604
0.29	26.15838	0.83	25.86615
0.26	26.1596	0.86	25.8796
0.23	26.15912	0.89	25.89875
0.2	26.15982	0.92	25.91328
0.17	26.1615	0.95	25.90415
0.14	26.162	0.98	25.8572
0.11	26.16403	1.01	25.73893
0.08	26.16417	1.04	25.48278
0.05	26.16552	1.07	24.88785
0.02	26.16698	1.1	23.47513
-0.01	26.16733	1.13	20.15132
-0.04	26.16748	1.16	13.5162

-0.07	26.16863	1.19	1.86649
-0.1	26.16932	1.22	-14.52814
-0.13	26.16953	1.25	-34.50455